

LEARNING OBJECTIVES

Competencies

- Illustrate the limit of a function using table of values and the graph of a function;
- Distinguish between $\lim_{x \to c} f(x)$ and $f(c)$;
- Illustrate the limit laws (theorems);
- Apply the limit laws in evaluating the limit of algebraic functions (i.e. polynomial, rational, and radical functions).

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INTRODUCTION

Limits are the backbone of calculus, and calculus is called the *Mathematics of Change*.

The study of limits is necessary in studying change in great detail. The evaluation of a particular limit is what underlies the formulation of the derivative and the antiderivative (indefinite integral) of a function.

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LIMIT OF A FUNCTION

Let f be a function of a single variable x. Consider a constant c which the variable x will approach (c may or may not be in the domain of f).

The *limit*, to be denoted by L, is the unique real value that $f(x)$ will approach as x approaches c. In symbols, we write this process as

$$\lim_{x \to c} f(x) = L$$

This is read, "The limit of $f(x)$ as x approaches c is L"

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Evaluating the Limit of a Function

USING TABLE OF VALUES

Consider $f(x) = x + 3$. Evaluate $\lim_{x \to 1} (x + 3)$

Left Hand Limit

x	$f(x)$
0	**3**
0.5	**3.5**
0.9	**3.9**
0.99	**3.99**
0.999	**3.999**
0.9999	**3.9999**
0.99999	**3.99999**

Right Hand Limit

x	$f(x)$
2	**5**
1.5	**4.5**
1.1	**4.1**
1.01	**4.01**
1.001	**4.001**
1.0001	**4.0001**
1.0001	**4.00001**

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Observe that as the values of x get closer and closer to $\mathbf{1}$, the values of $f(x)$ get closer and closer to $\mathbf{4}$.

This behavior can be shown no matter what set of values, or what direction, is taken in approaching 1.

In symbols,

$$\lim_{x \to 1} (x + 3) = 4$$

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EXAMPLE 2

Use table of values to evaluate $\lim_{x \to 0} |x|$

Left Hand Limit

x	$f(x)$
-1	
-0.5	
-0.1	
-0.01	
-0.001	
-0.0001	
-0.00001	

Right Hand Limit

x	$f(x)$
1	
0.5	
0.1	
0.01	
0.001	
0.0001	
0.00001	

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Solution:

Left Hand Limit

x	$f(x)$
-1	**1**
-0.5	**0.5**
-0.1	**0.1**
-0.01	**0.01**
-0.001	**0.001**
-0.0001	**0.0001**
-0.00001	**0.00001**

$$\lim_{x \to 0^-} |x| = 0$$

Right Hand Limit

x	$f(x)$
1	**1**
0.5	**0.5**
0.1	**0.1**
0.01	**0.01**
0.001	**0.001**
0.0001	**0.0001**
0.00001	**0.00001**

$$\lim_{x \to 0^+} |x| = 0$$

Thus, $\lim_{x \to 0} |x| = 0$

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EXAMPLE 3

Use xy-table to evaluate $\lim_{x \to 1} \dfrac{x^2 - 2x + 1}{x - 1}$

Left Hand Limit

x	$f(x)$
0	**0**
0.5	**-0.5**
0.9	**-0.1**
0.99	**-0.01**
0.999	**-0.001**
0.9999	**-0.0001**
0.99999	**-0.00001**

Right Hand Limit

x	$f(x)$
2	**1**
1.5	**0.5**
1.1	**0.1**
1.01	**0.01**
1.001	**0.001**
1.0001	**0.0001**
1.00001	**0.00001**

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EXAMPLE 4

Evaluate $\lim_{x \to 3} f(x)$ if $f(x) = \begin{cases} x + 1, & \text{if } x \geq 3 \\ 5 - x, & \text{if } x < 3 \end{cases}$

Left Hand Limit

x	$f(x)$
2	**3**
2.5	**2.5**
2.9	**2.1**
2.99	**2.01**
2.999	**2.001**
2.9999	**2.0001**
2.99999	**2.00001**

Right Hand Limit

x	$f(x)$
4	**5**
3.5	**4.5**
3.1	**4.1**
3.01	**4.01**
3.001	**4.001**
3.0001	**4.0001**
3.00001	**4.00001**

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ONE-SIDED LIMITS

If x approaches c from the left, or through values less than c, then

$$\lim_{x \to c^-} f(x) = L_1$$

If x approaches c from the right, or through values greater than c, then

$$\lim_{x \to c^+} f(x) = L_2$$

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TWO-SIDED LIMITS

Furthermore, we say that

$$\lim_{x \to c} f(x) = L$$

if and only if

$$\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$$

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TWO-SIDED LIMITS

In other words, for a limit L to exist, the limits from the left and from the right must both exist and equal. Therefore

$$\lim_{x \to c} f(x) \text{ DNE}$$

whenever

$$\lim_{x \to c^-} f(x) \neq \lim_{x \to c^+} f(x)$$

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Evaluating the Limit of a Function

USING GRAPHICAL METHOD

Evaluate:

$$\lim_{x \to 1} f(x)$$

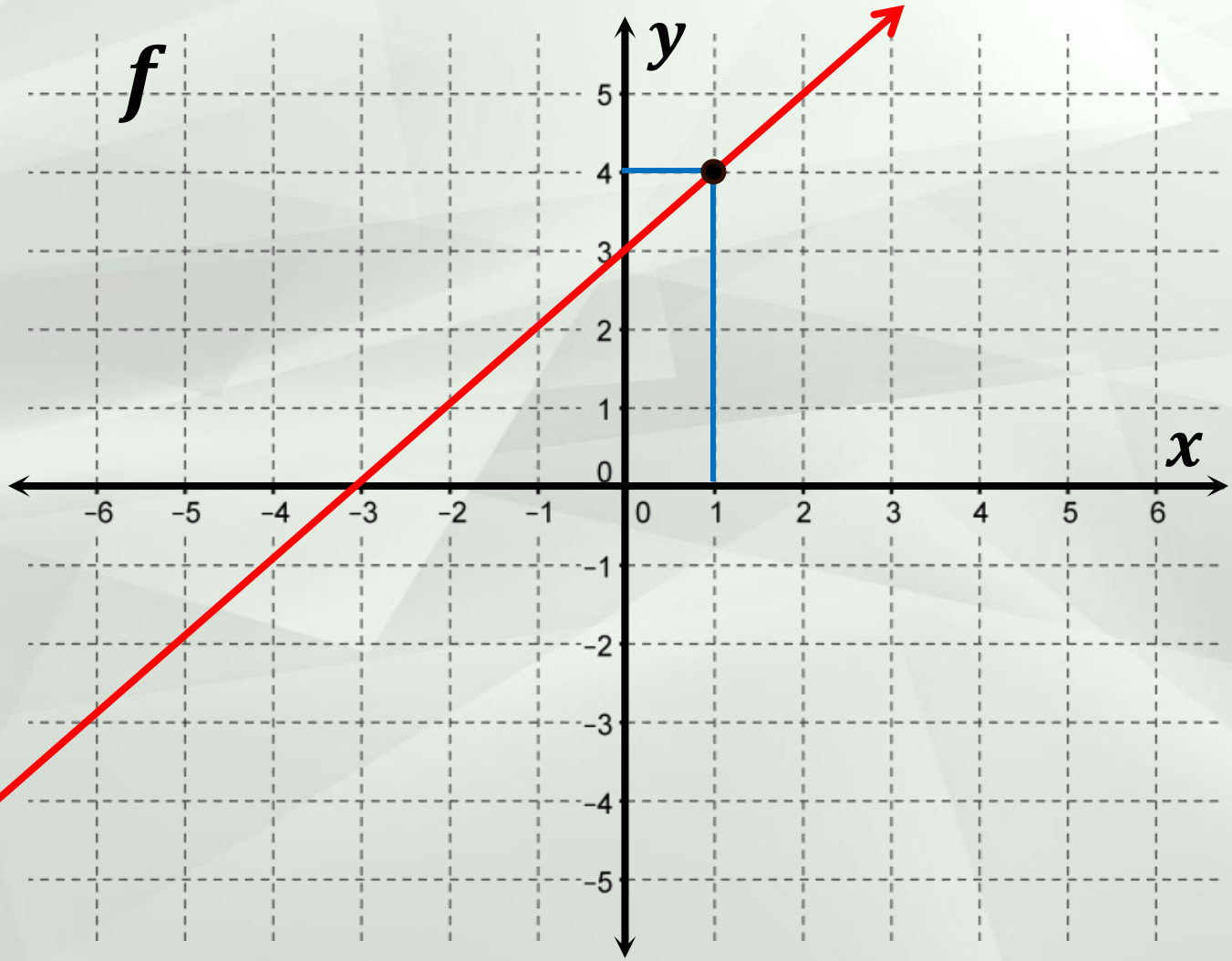
Since

$$\lim_{x \to 1^-} f(x) = 4$$

$$\lim_{x \to 1^+} f(x) = 4$$

Then,

$$\lim_{x \to 1} f(x) = 4$$



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Evaluating the Limit of a Function

USING GRAPHICAL METHOD

Evaluate:

$$\lim_{x \to 4} f(x)$$

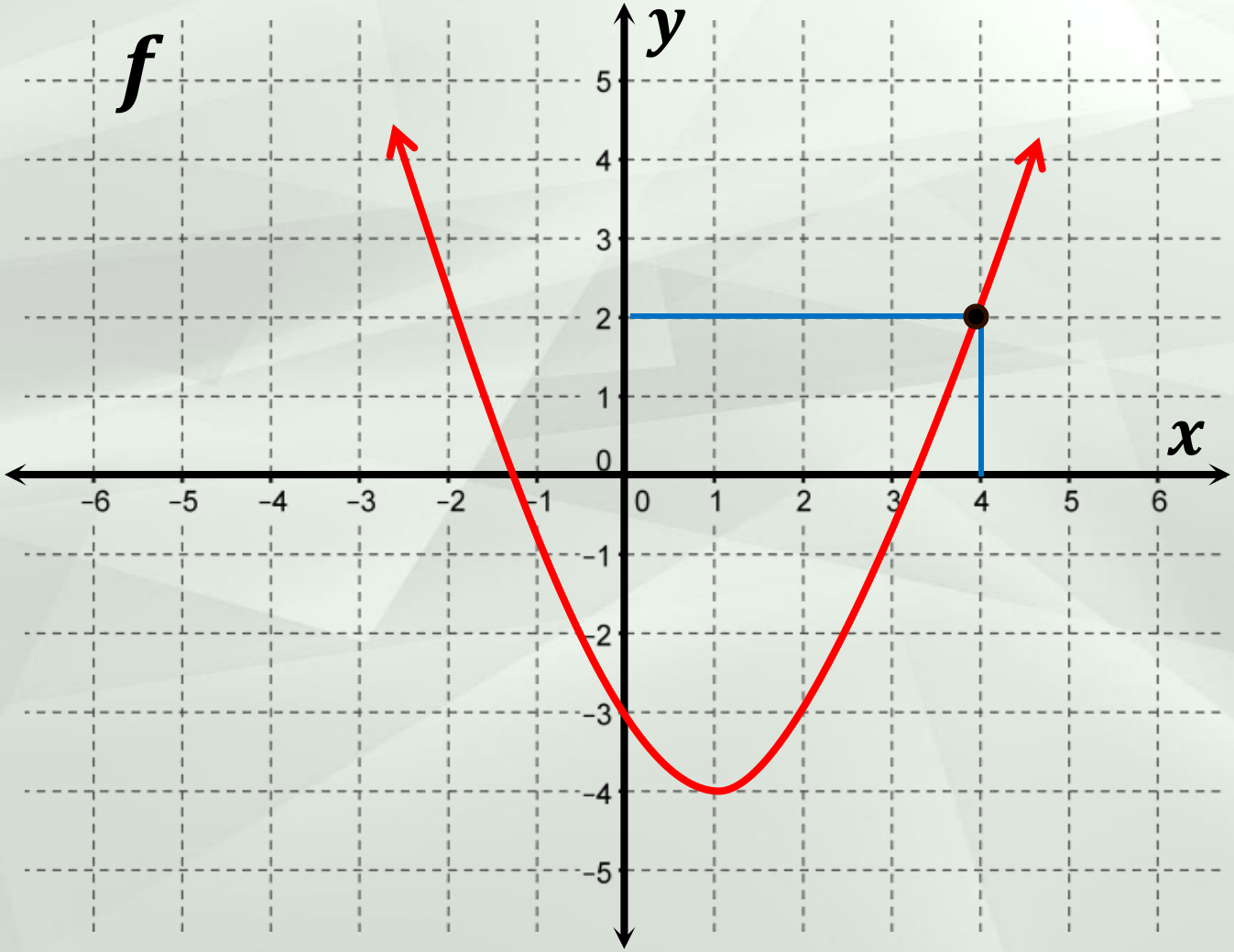
Since

$$\lim_{x \to 4^-} f(x) = 2$$

$$\lim_{x \to 4^+} f(x) = 2$$

Then,

$$\lim_{x \to 4} f(x) = 2$$



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Evaluating the Limit of a Function

USING GRAPHICAL METHOD

Evaluate:

$$\lim_{x \to 2} g(x)$$

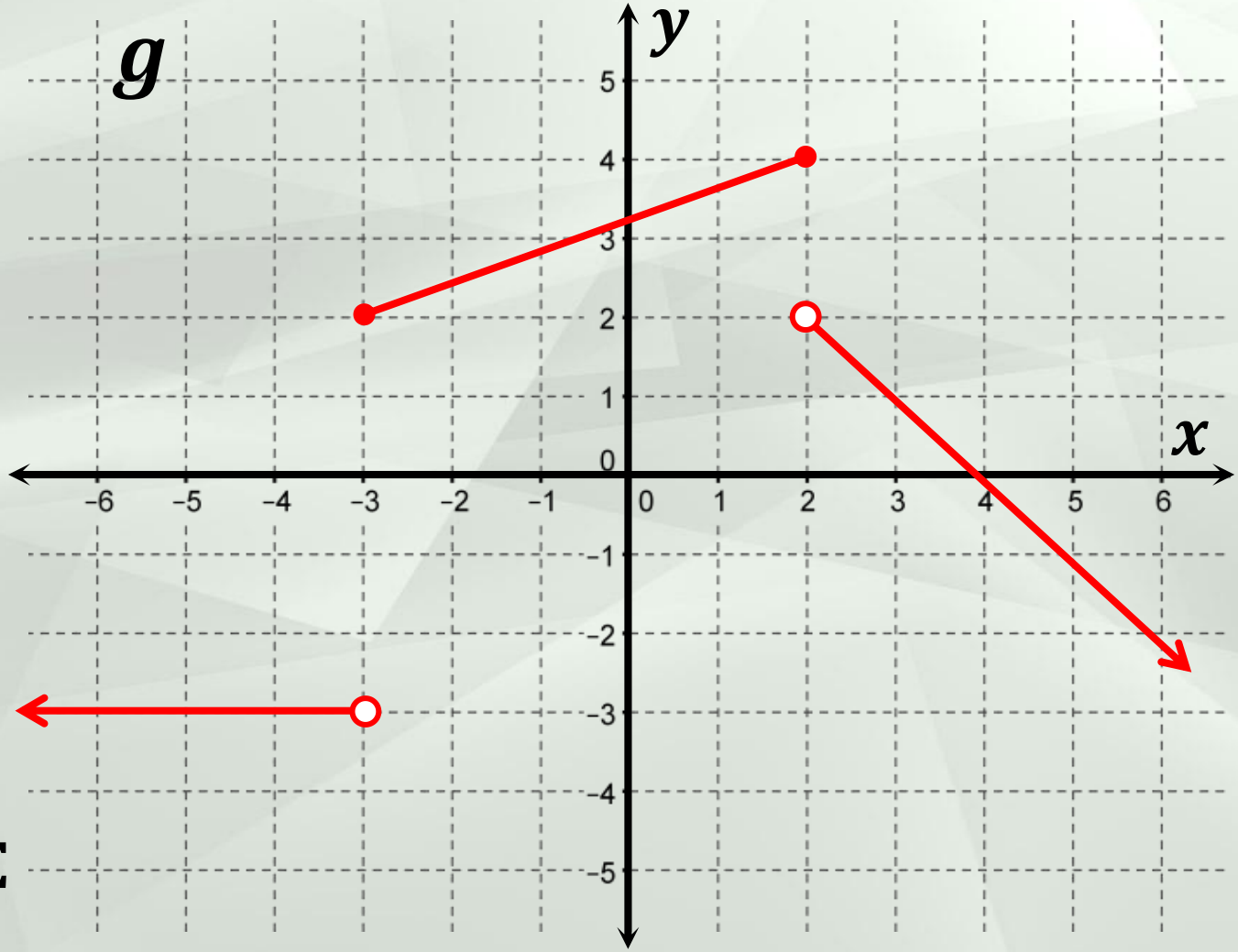
Since

$$\lim_{x \to 2^-} g(x) = 4$$

$$\lim_{x \to 2^+} g(x) = 2$$

Then,

$$\lim_{x \to 2} g(x) \text{ DNE}$$



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Evaluating the Limit of a Function

USING GRAPHICAL METHOD

Evaluate:

$$\lim_{x \to 3} h(x)$$

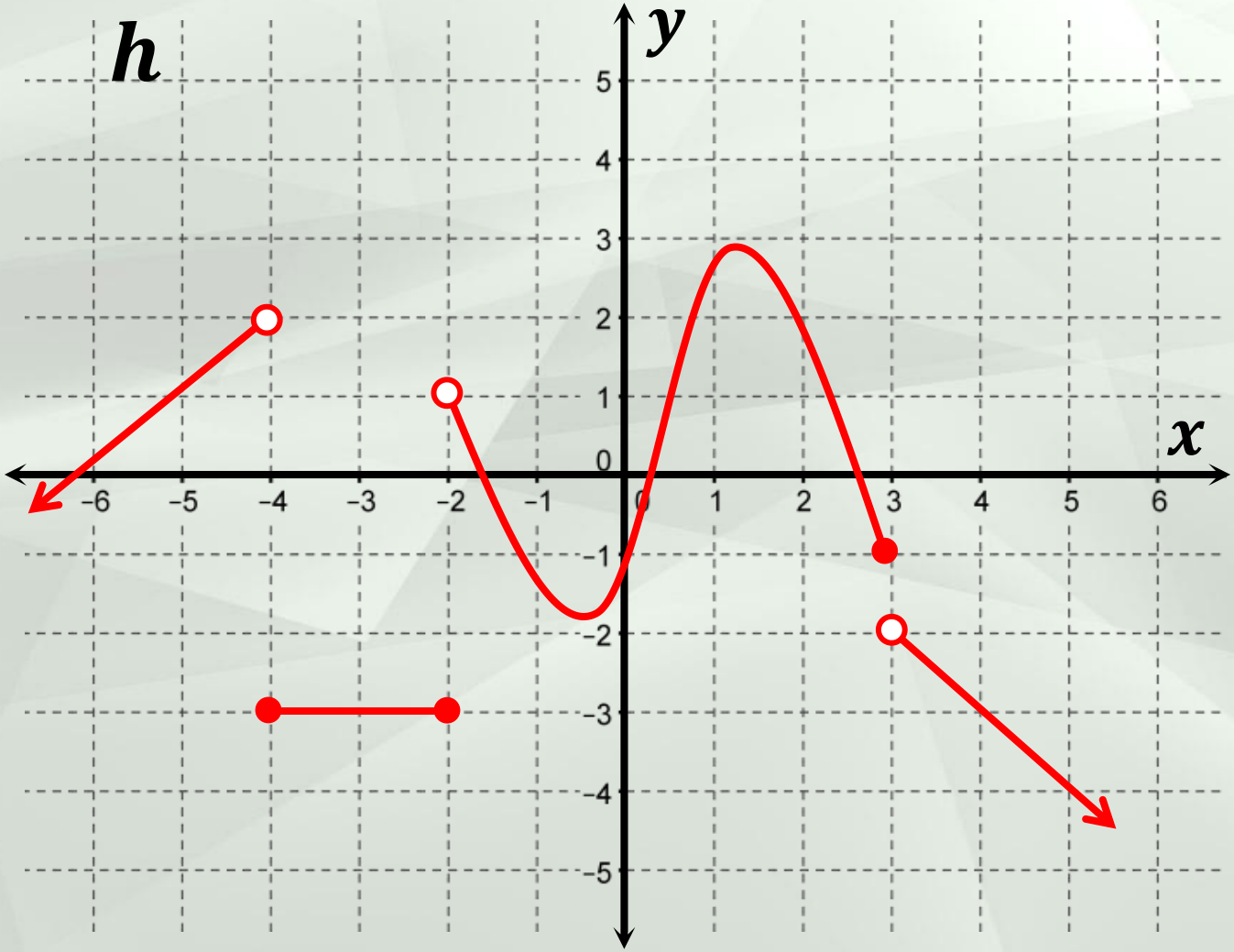
Since

$$\lim_{x \to 3^-} h(x) = -1$$

$$\lim_{x \to 3^+} h(x) = -2$$

Then,

$$\lim_{x \to 3} h(x) \text{ DNE}$$



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EXERCISES

Construct a table of values to investigate and evaluate the following limits.

1. $\lim_{x \to 1} 2x + 3$

2. $\lim_{x \to 2} x^2 - 1$

3. $\lim_{x \to 3} \frac{10}{x - 2}$

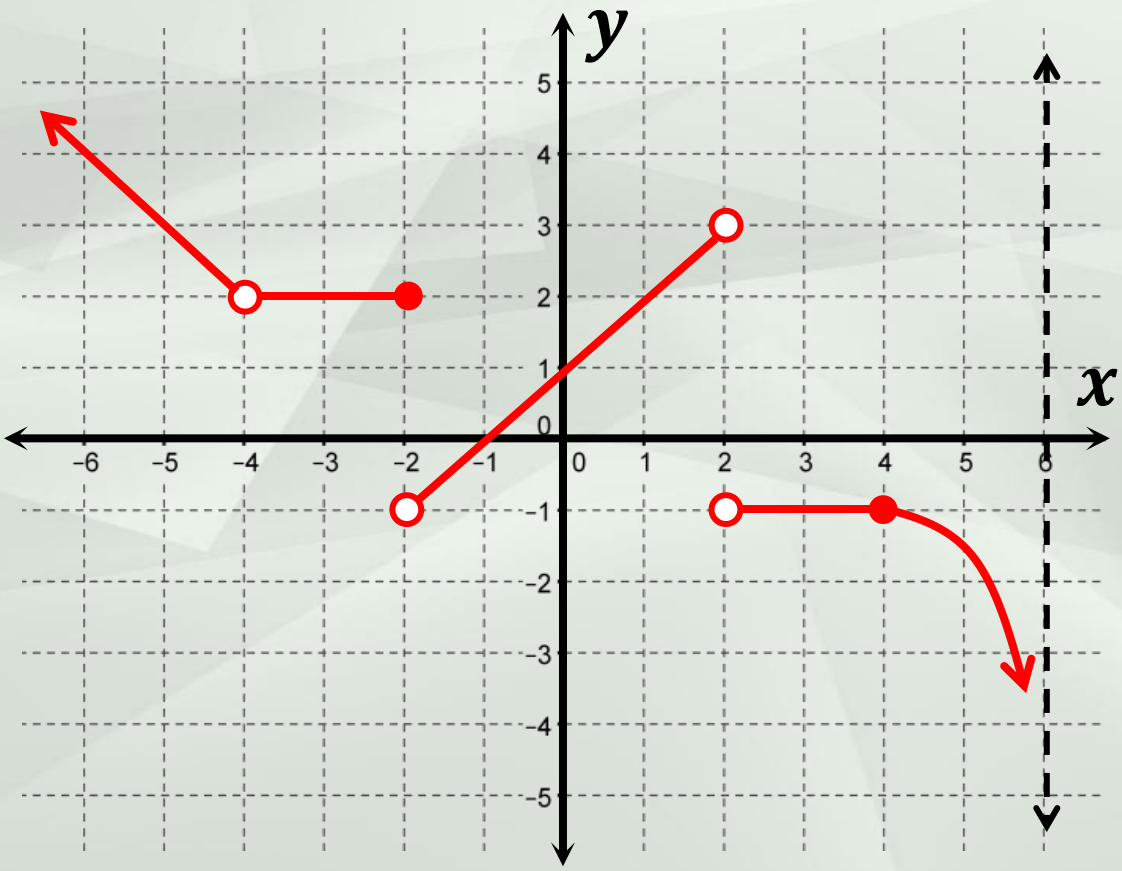
4. $\lim_{x \to 4} f(x)$ if $f(x) = \begin{cases} x + 3, \text{ if } x \geq 4 \\ x - 1, \text{ if } x < 4 \end{cases}$

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EXERCISES

Evaluate each limit (if it exists) using the graph below.

1. $\lim_{x \to -1} f(x)$
2. $\lim_{x \to -4} f(x)$
3. $\lim_{x \to 2} f(x)$
4. $\lim_{x \to -2} f(x)$
5. $\lim_{x \to 4} f(x)$



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THE LIMIT OF A FUNCTION at c VERSUS THE VALUE OF THE FUNCTION AT c

Critical to the study of limits is the understanding that the value of

$$\lim_{x \to c} f(x)$$

may be distinct from the value of the function at $x = c$, that is, $f(c)$.

Generally, $\lim_{x \to c} f(x)$ has nothing to do with $f(c)$!

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The LIMIT vs the VALUE of f

Recall that $\lim_{x \to 1} (x + 3) = 4$

Left Hand Limit

x	$f(x)$
0	**3**
0.5	**3.5**
0.9	**3.9**
0.99	**3.99**
0.999	**3.999**
0.9999	**3.9999**
0.99999	**3.99999**

Right Hand Limit

x	$f(x)$
2	**5**
1.5	**4.5**
1.1	**4.1**
1.01	**4.01**
1.001	**4.001**
1.0001	**4.0001**
1.0001	**4.00001**

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The LIMIT vs the VALUE of f

Note that the value of the function at $x = 1$ is also equal to 4 or $\boldsymbol{f(1) = 4}$.

However, it's not always the case. Consider following function

$$\boldsymbol{f(x) = \begin{cases} x + 1, \text{ if } x \geq 3 \\ 5 - x, \text{ if } x < 3 \end{cases}}$$

Then, $\boldsymbol{f(3) = 4}$ and $\boldsymbol{\lim_{x \to 3} f(x)}$ **DNE**

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The LIMIT vs the VALUE of f

Example:

Given that

$$f(x) = \begin{cases} 5x + 4, \text{ if } x \neq 1 \\ 1 - 3x, \text{ if } x = 1 \end{cases}$$

Calculate $f(1)$ and $\lim_{x \to 1} f(x)$

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The LIMIT vs the VALUE of f

$$f(x) = \begin{cases} 5x + 4, \text{ if } x \neq 1 \\ 1 - 3x, \text{ if } x = 1 \end{cases}$$

Left Hand Limit

x	$f(x)$
0	**4**
0.5	**6.5**
0.9	**8.5**
0.99	**8.95**
0.999	**8.995**
0.9999	**8.9995**
0.99999	**8.99995**

Right Hand Limit

x	$f(x)$
2	**14**
1.5	**11.5**
1.1	**9.5**
1.01	**9.05**
1.001	**9.005**
1.0001	**9.0005**
1.00001	**9.00005**

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The LIMIT vs the VALUE of f

Example:

Given that

$$f(x) = \begin{cases} 5x + 4, \text{ if } x \neq 1 \\ 1 - 3x, \text{ if } x = 1 \end{cases}$$

$f(1) = -2$ and $\lim_{x \to 1} f(x) = 9$

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BASIC LIMIT THEOREMS

1. **Constant Rule**: The limit of a constant is itself. If k is any constant, then

$$\lim_{x \to c} k = k$$

Examples:

$$\lim_{x \to 1} 3.5 = 3.5$$

$$\lim_{x \to 1} 2 \cdot 5^{25} = 2 \cdot 5^{25}$$

$$\lim_{x \to \frac{1}{2}} \pi = \pi$$

$$\lim_{x \to 2019} 2018 = 2018$$

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BASIC LIMIT THEOREMS

2. **Identity Rule**: The limit of x as x approaches c is equal to c.

$$\lim_{x \to c} x = c$$

Examples:

$$\lim_{x \to 5} x = 5$$

$$\lim_{x \to -21} x = -21$$

$$\lim_{x \to \frac{2}{3}} x = \frac{2}{3}$$

$$\lim_{x \to 11} x = 11$$

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BASIC LIMIT THEOREMS

3. **Constant Multiple Rule**: The limit of a multiple of a function is the multiple of the limit of the function.

If k is a constant and $\lim_{x \to c} f(x) = L$ then

$$\lim_{x \to c} k \cdot f(x) = k \cdot \lim_{x \to c} f(x) = k \cdot L$$

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For example: Let $\lim_{x \to c} f(x) = \mathbf{4}$

1. $\lim_{x \to c} 3 \cdot f(x) = 3 \cdot \lim_{x \to c} f(x) = 3 \cdot 4 = \mathbf{12}$

2. $\lim_{x \to c} 8 \cdot f(x) = 8 \cdot \lim_{x \to c} f(x) = 8 \cdot 4 = \mathbf{32}$

3. $\lim_{x \to c} -1 \cdot f(x) = -1 \cdot \lim_{x \to c} f(x) = -1 \cdot 4 = \mathbf{-4}$

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- $\lim_{x \to 2} 4x = 4 \times \lim_{x \to 2} x = 4 \times 2 = 8$
- $\lim_{x \to 3} x^2 = \lim_{x \to 3} (x \times x) = \lim_{x \to 3} x \times \lim_{x \to 3} x = 3 \times 3 = 9$
- $\lim_{x \to 4} x^3 = (\lim_{x \to 4} x)\text{^}3 = 4^3 = 64$

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BASIC LIMIT THEOREMS

4. **Addition Rule**: The limit of a sum of functions is the sum of the limits of the individual functions.

Suppose $\lim_{x \to c} f(x) = L$ & $\lim_{x \to c} g(x) = M$

$$\lim_{x \to c} [f(x) + g(x)] = \lim_{x \to c} f(x) + \lim_{x \to c} g(x)$$
$$= L + M$$

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BASIC LIMIT THEOREMS

5. **Subtraction Rule**: The limit of a difference of functions is the difference of the limits of the individual functions.

Suppose $\lim_{x \to c} f(x) = L$ & $\lim_{x \to c} g(x) = M$

$$\lim_{x \to c} [f(x) - g(x)] = \lim_{x \to c} f(x) - \lim_{x \to c} g(x)$$
$$= L - M$$

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For example: Let

$$\lim_{x \to c} f(x) = \mathbf{3} \text{ and } \lim_{x \to c} g(x) = -\mathbf{2}$$

1. $\lim_{x \to c} f(x) + \lim_{x \to c} g(x) = 3 + (-2) = \mathbf{1}$

2. $\lim_{x \to c} f(x) - \lim_{x \to c} g(x) = 3 - (-2) = \mathbf{5}$

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BASIC LIMIT THEOREMS

6. **Multiplication Rule**: The limit of a product of functions is equal to the product of their limits.

Suppose $\lim_{x \to c} f(x) = L$ & $\lim_{x \to c} g(x) = M$

$$\lim_{x \to c} [f(x) \cdot g(x)] = \lim_{x \to c} f(x) \cdot \lim_{x \to c} g(x)$$
$$= L \cdot M$$

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For example: Let

$$\lim_{x \to c} f(x) = \mathbf{5} \text{ and } \lim_{x \to c} g(x) = -\mathbf{2}$$

1. $$\lim_{x \to c} f(x) \cdot \lim_{x \to c} g(x) = 5 \cdot (-2) = -\mathbf{10}$$

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BASIC LIMIT THEOREMS

7. **Division Rule**: The limit of a quotient of functions is equal to the quotient of their limits, provided the denominator limit is not equal to 0.

Suppose $\lim_{x \to c} f(x) = L$ & $\lim_{x \to c} g(x) = M$

$$\lim_{x \to c} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)} = \frac{L}{M}, \; M \neq 0$$

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For example:

1. If $\lim_{x \to c} f(x) = \mathbf{10}$ and $\lim_{x \to c} g(x) = \mathbf{-5}$

$$\lim_{x \to c} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)} = \frac{\mathbf{10}}{\mathbf{-5}} = \mathbf{-2}$$

2. If $\lim_{x \to c} f(x) = \mathbf{11}$ and $\lim_{x \to c} g(x) = \mathbf{0}$

$$\lim_{x \to c} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)} = \frac{\mathbf{11}}{\mathbf{0}} \ \mathbf{DNE}$$

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BASIC LIMIT THEOREMS

8. **Power Rule**: The limit of an integer power p of a function is just the power of the limit of the function.

Let $\lim_{x \to c} f(x) = L$

$$\lim_{x \to c} (f(x))^p = (\lim_{x \to c} f(x))^p = L^p$$

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For example:

1. If $\lim_{x \to c} f(x) = -2$

$$\lim_{x \to c} (f(x))^3 = (\lim_{x \to c} f(x))^3 = (-2)^3 = -8$$

2. If $\lim_{x \to c} f(x) = \frac{1}{2}$

$$\lim_{x \to c} (f(x))^{-2} = (\lim_{x \to c} f(x))^{-2} = \left(\frac{1}{2}\right)^{-2} = 4$$

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For example:

1. If $\lim_{x \to c} f(x) = -2$

$$\lim_{x \to c} (f(x))^3 = (\lim_{x \to c} f(x))^3 = (-2)^3 = -8$$

2. If $\lim_{x \to c} f(x) = \frac{1}{2}$

$$\lim_{x \to c} (f(x))^{-2} = (\lim_{x \to c} f(x))^3 = \left(\frac{1}{2}\right)^{-2} = 4$$

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BASIC LIMIT THEOREMS

9. **Radical Rule**: If n is a positive integer, the limit of the nth root of a function is just the nth root of the limit of the function, provided that the nth root of the limit is a real number.

If $\lim_{x \to c} f(x) = L$, $L > 0$ if n is even then

$$\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to c} f(x)} = \sqrt[n]{L}$$

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For example:

1. If $\lim_{x \to c} f(x) = \mathbf{100}$

$$\lim_{x \to c} \sqrt{f(x)} = \sqrt{\lim_{x \to c} f(x)} = \sqrt{100} = \mathbf{10}$$

2. If $\lim_{x \to c} f(x) = \mathbf{-1}$

$$\lim_{x \to c} \sqrt{f(x)} = \sqrt{\lim_{x \to c} f(x)} = \sqrt{-1} \quad \mathbf{DNE}$$

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PRACTICE EXERCISES

Assume the following

$$\lim_{x\to c} f(x) = \frac{1}{2},\ \lim_{x\to c} g(x) = -4\ \&\ \lim_{x\to c} h(x) = 8$$

Compute the following limits:

a. $\lim_{x\to c} [10 \cdot f(x)]$

b. $\lim_{x\to c} \sqrt{2 \cdot f(x)}$

c. $\lim_{x\to c} [f(x) + h(x)]$

d. $\lim_{x\to c} [3f(x) - 2g(x)]$

e. $\lim_{x\to c} [g(x) \cdot h(x)]$

f. $\lim_{x\to c} [f(x) - g(x)]^2$

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PRACTICE EXERCISES

A. Assume the following

$$\lim_{x\to c} f(x) = \frac{2}{5},\ \lim_{x\to c} g(x) = 20\ \&\ \lim_{x\to c} h(x) = -1$$

Compute the following limits:

g. $\lim_{x\to c} [f(x) - \frac{3}{5} g(x)]$

h. $\lim_{x\to c} (h(x) \cdot g(x))^2$

i. $\lim_{x\to c} \left[\frac{f(x)}{g(x)}\right]$

j. $\lim_{x\to c} \left[\frac{15 \cdot f(x)}{(h(x))^2}\right]$

k. $\lim_{x\to c} \sqrt[3]{50 \cdot g(x)}$

l. $\lim_{x\to c} \left[\frac{\sqrt{g(x) - 11}}{(5 \cdot f(x))^3}\right]$

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B. Determine whether the statement is always True or False. If it is false explain what makes it false, or provide a counterexample.

1. If $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ both exist, then $\lim_{x\to c} f(x) \pm \lim_{x\to c} g(x)$ always exists.

2. If $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ both exist, then $\lim_{x\to c} f(x) \cdot \lim_{x\to c} g(x)$ always exists.

3. If $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ both exist, then $\lim_{x\to c} \frac{f(x)}{g(x)}$ always exists.

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4. If $\lim_{x \to c} f(x)$ exists and p is an integer, then $\lim_{x \to c} (f(x))^p$ always exists.

5. If $\lim_{x \to c} f(x)$ exists and n is a real number, then $\lim_{x \to c} \sqrt[n]{f(x)}$ always exists.

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LIMITS OF POLYNOMIAL, RATIONAL AND RADICAL FUNCTIONS

Theorem 2

Let f be a polynomial function of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_1 x + a_0$$

If c is a real number, then

$$\lim_{x \to c} f(x) = f(c)$$

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SAMPLE PROBLEM

Evaluate:

$$\lim_{x \to 2} (2x^3 - 5x^2 + 4x - 7)$$

Solution:

$$f(2) = 2(2)^3 - 5(2)^2 + 4(2) - 7$$

$$= 2(8) - 5(4) + 4(2) - 7$$

$$= 16 - 20 + 8 - 7 = \mathbf{-3}$$

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TRY THIS!

Evaluate:

$$\lim_{x \to 1} (5x^6 - 3x^5 + 5x^3 - 2x + 11)$$

$$\lim_{x \to -1} (2016x^{2017} - 2018x^{2019})$$

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LIMITS OF POLYNOMIAL, RATIONAL AND RADICAL FUNCTIONS

Theorem 3

Let h be a rational function of the form

$$h(x) = \frac{f(x)}{g(x)}$$

where f and g are polynomial functions. If c is a real number and $g(c) \neq 0$, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{f(c)}{g(c)}$$

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SAMPLE PROBLEM

Evaluate:

$$\lim_{x \to 1} \frac{2x + 13}{x^2 - 3x + 5}$$

Solution:

$$f(1) = \frac{2(1) + 13}{(1)^2 - 3(1) + 5} = \frac{15}{3} = \mathbf{5}$$

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TRY THIS!

Evaluate:

$$\lim_{x \to -1} \frac{2x^2 - 3x + 7}{x^3 - 3x^2 + 4x - 10}$$

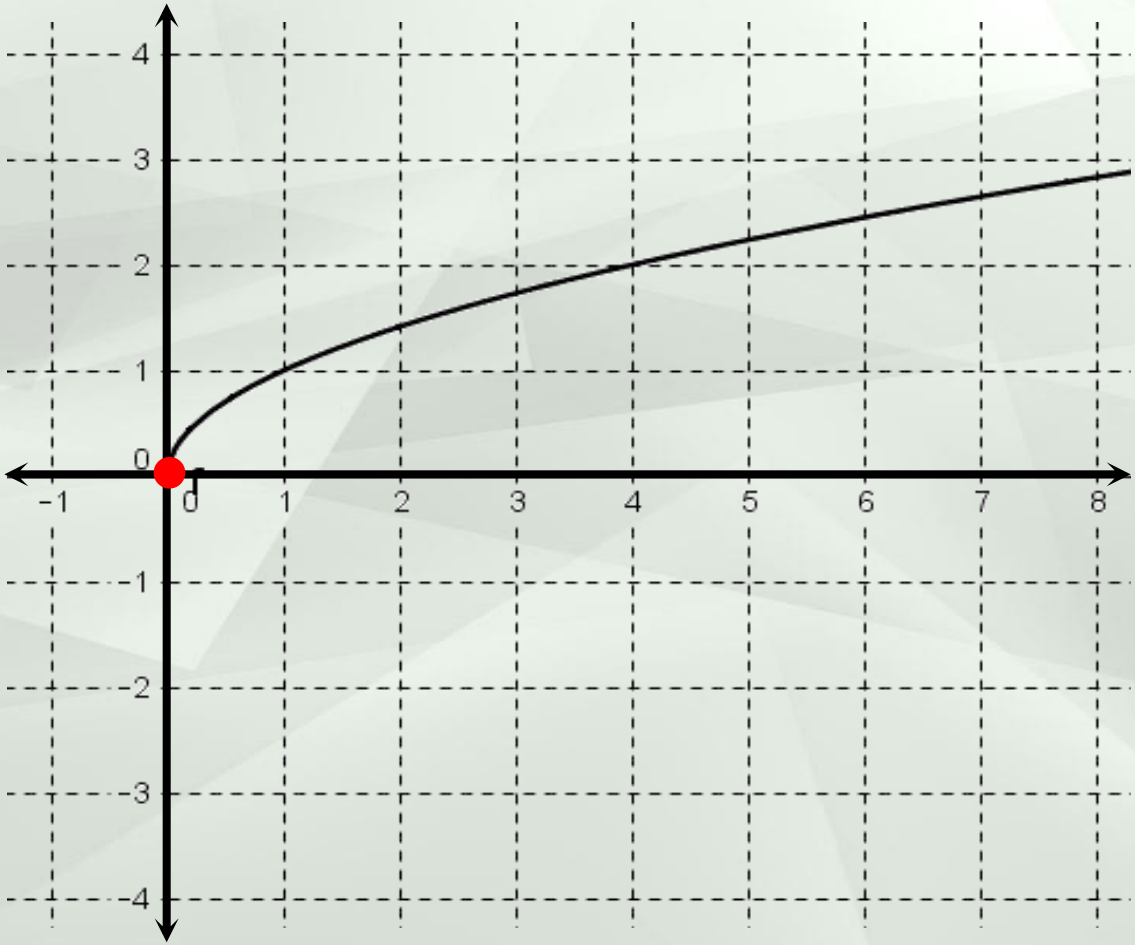
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SAMPLE PROBLEM

Let $f(x) = \sqrt{x}$

Evaluate: $f(0)$

and $\lim_{x \to 0} f(x)$



$f(0) = \mathbf{0}$

$\lim_{x \to 0} \sqrt{x}$ **DNE**

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SAMPLE PROBLEM

Evaluate:

$$\lim_{x \to 3} \sqrt{x+1}$$

Solution:

$$\lim_{x \to 3} \sqrt{x+1} = \sqrt{\lim_{x \to 3}(x+1)} = \sqrt{3+1} = \mathbf{2}$$

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SAMPLE PROBLEM

Evaluate:

$$\lim_{x \to 1} \frac{\sqrt{x^2 + 8}}{x^2 + 4x - 2}$$

Solution:

$$\lim_{x \to 1} \frac{\sqrt{x^2 + 8}}{x^2 + 4x - 2} = \frac{\sqrt{(1)^2 + 8}}{(1)^2 + 4(1) - 2} = \frac{\sqrt{9}}{3} = \mathbf{1}$$

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INDETERMINATE FORM

Indeterminate Form of Type $\frac{"0"}{0}$

If $\lim_{x \to c} f(x) = 0$ and $\lim_{x \to c} g(x) = 0$, then $\lim_{x \to c} \frac{f(x)}{g(x)}$ is called an indeterminate form of type $\frac{"0"}{0}$.

Note: Indeterminate form is never a final answer!

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SAMPLE PROBLEM

Evaluate: $\lim_{x \to 0} \dfrac{2x^2 - 5x}{x}$

Solution:

$$\lim_{x \to 0} \frac{2x^2 - 5x}{x} = \frac{2(0)^2 - 5(0)}{0} = \frac{0}{0}$$

$$\lim_{x \to 0} \frac{\cancel{x}(2x - 5)}{\cancel{x}} = \lim_{x \to 0}(2x - 5) = 2(0) - 5 = -\mathbf{5}$$

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SAMPLE PROBLEM

Evaluate: $\lim_{x \to 1} \dfrac{x^2 - 2x + 1}{x - 1}$

Solution:

$$\lim_{x \to 1} \frac{x^2 - 2x + 1}{x - 1} = \frac{1^2 - 2(1) + 1}{1 - 1} = \frac{0}{0}$$

$$\lim_{x \to 1} \frac{(x - 1)(x - 1)}{x - 1} = \lim_{x \to 1}(x - 1) = 1 - 1 = 0$$

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SAMPLE PROBLEM

Evaluate: $\lim_{x \to 2} \dfrac{x^2 - x - 2}{x^2 - 4}$

Solution:

$$\lim_{x \to 2} \frac{(x-2)(x+1)}{(x+2)(x-2)} = \lim_{x \to 2} \frac{x+1}{x+2} = \frac{2+1}{2+2} = \frac{\mathbf{3}}{\mathbf{4}}$$

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- $p(x) = x^3 - 8$
- $p(2) = 2^3 - 8 = 0$
- $x - 2$ is a factor

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SAMPLE PROBLEM

Evaluate: $\lim_{x \to 2} \dfrac{x^3 - 8}{2 - x}$

Solution:

$$\lim_{x \to 2} \frac{(x-2)(x^2+2x+4)}{-(x-2)} = \lim_{x \to 2} \frac{x^2+2x+4}{-1}$$

$$= \frac{2^2 + 2(2) + 4}{-1} = \frac{12}{-1} = \mathbf{-12}$$

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SAMPLE PROBLEM

Evaluate: $\lim_{x \to 1} \dfrac{x-1}{\sqrt{x}-1}$

Solution:

$$\lim_{x \to 1} \frac{x-1}{\sqrt{x}-1} \cdot \frac{\sqrt{x}+1}{\sqrt{x}+1} = \lim_{x \to 1} \frac{(x-1)(\sqrt{x}+1)}{x-1}$$

$$\lim_{x \to 1} \sqrt{x}+1 = \sqrt{1}+1 = \mathbf{2}$$

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SAMPLE PROBLEM

Evaluate: $\lim_{x\to 4} \dfrac{x^2+x-20}{2-\sqrt{x}}$

Solution:

$$\lim_{x\to 4} \frac{x^2+x-20}{2-\sqrt{x}} \cdot \frac{2+\sqrt{x}}{2+\sqrt{x}} = \lim_{x\to 4} \frac{(x-4)(x+5)(2+\sqrt{x})}{-(x-4)}$$

$$\lim_{x\to 4} \frac{(x+5)\left(2+\sqrt{x}\right)}{-1} = \frac{(4+5)\left(2+\sqrt{4}\right)}{-1} = \frac{(9)(4)}{-1} = -\mathbf{36}$$

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PRACTICE EXERCISES

Evaluate each limit (if it exists).

1. $\lim_{x \to \frac{1}{2}} (x^2 + 10x - 2)$
2. $\lim_{x \to -3} (x^3 - 2x^2 + 8x - 1)$
3. $\lim_{x \to 1} \left(\frac{4x - 1}{x + 2} \right)$
4. $\lim_{x \to 0} \left(\frac{3x - 10}{x^2 - 3x + 5} \right)$
5. $\lim_{x \to -2} \left(\sqrt{x + 11} \right)$
6. $\lim_{x \to 0} \left(\frac{x^2 - 8x}{2x} \right)$
7. $\lim_{x \to 3} \left(\frac{x^2 - 9}{x - 3} \right)$
8. $\lim_{x \to 2} \left(\frac{2x^3 - 16}{x^2 - 4} \right)$
9. $\lim_{x \to 1} \left(\frac{1 - \sqrt{x}}{x - 1} \right)$
10. $\lim_{x \to 25} \left(\frac{2x - 50}{5 - \sqrt{x}} \right)$

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INFINITE LIMITS

Consider the function $f(x) = \frac{1}{x^2}$. Note that the function is not defined at $x = 0$ but we can describe the behavior of f as x approaches 0.

	x	$f(x)$
Left Hand Limit	-1	**1**
	-0.5	**4**
	-0.25	**16**
	-0.1	**100**
	-0.01	**10,000**
	-0.001	**1,000,000**
	-0.0001	**100,000,000**

$$\lim_{x \to 0^-} f(x) = +\infty$$

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INFINITE LIMITS

Similarly, if we let x approach 0 through values greater than 0 or from the right, we observe that the $f(x)$ values get larger and larger without bound.

$$\lim_{x \to 0^+} f(x) = +\infty$$

	x	$f(x)$
	1	**1**
	0.5	**4**
	0.25	**16**
Left Hand Limit	0.1	**100**
	0.01	**10,000**
	0.001	**1,000,000**
	0.0001	**100,000,000**

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Since $\lim_{x \to 0^-} f(x) = +\infty$ and $\lim_{x \to 0^+} f(x) = +\infty$

We may conclude that $\lim_{x \to 0} f(x) = +\infty$

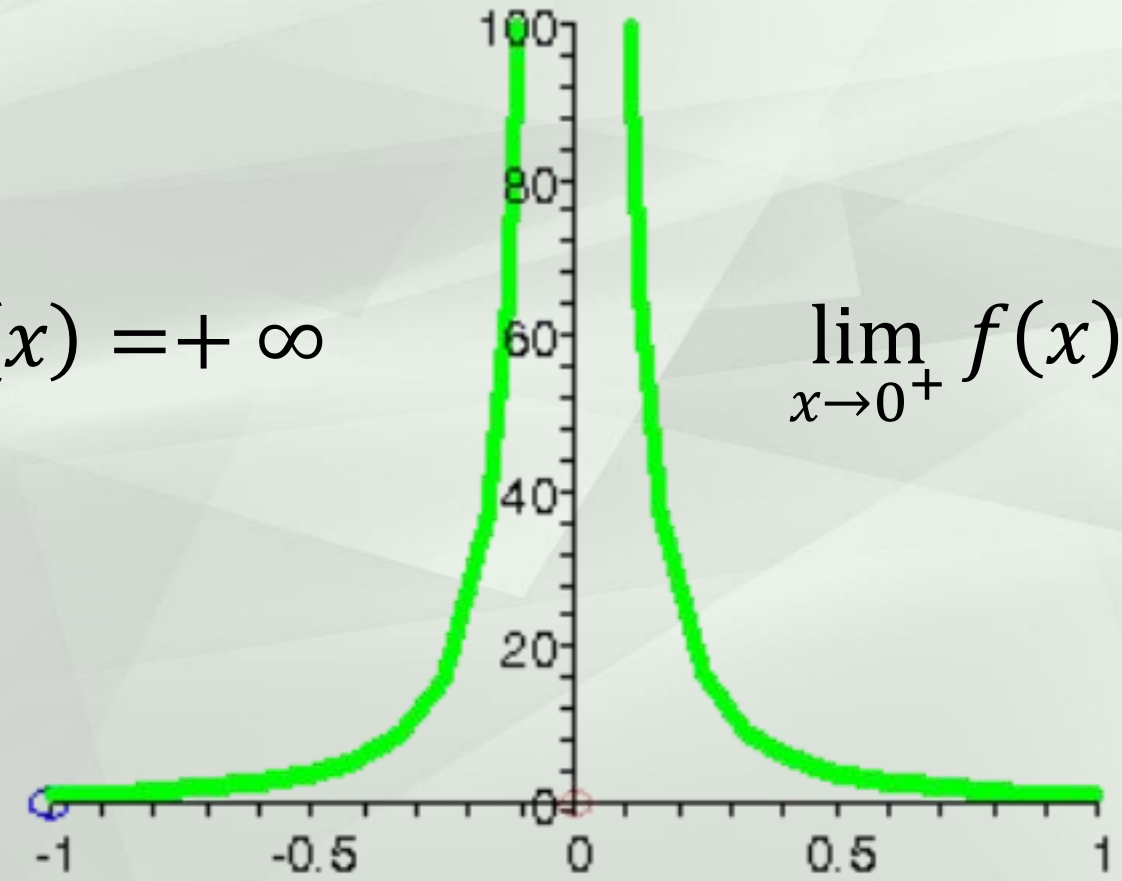
Note:

The symbol ∞ is NOT a number. In fact, it does not hold a specific value. So, $\lim_{x \to c} f(x) = \pm\infty$ only describes the behavior of the function near $x = c$, but it **does not exist** as a real number.

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$$\lim_{x \to 0} f(x) = +\infty$$

$$\lim_{x \to 0^-} f(x) = +\infty$$



$$\lim_{x \to 0^+} f(x) = +\infty$$

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INFINITE LIMITS

Definition

The limit of $f(x)$ as x approaches c is *positive infinity*, denoted by

$$\lim_{x \to c} f(x) = +\infty$$

if the value of $f(x)$ increases without bound whenever the values of x get closer and closer to c.

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INFINITE LIMITS

Definition

The limit of $f(x)$ as x approaches c is *negative infinity*, denoted by

$$\lim_{x \to c} f(x) = -\infty$$

if the value of $f(x)$ decreases without bound whenever the values of x get closer and closer to c.

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ONE-SIDED INFINITE LIMITS

Theorem

If n is any positive integer, then

$$\lim_{x \to 0^+} \frac{1}{x^n} = +\infty$$

$$\lim_{x \to 0^-} \frac{1}{x^n} = \begin{cases} +\infty & \text{if } n \text{ is even} \\ -\infty & \text{if } n \text{ is odd} \end{cases}$$

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EXAMPLES

Evaluate the following infinite limits

1. $\lim_{x \to 0^+} \frac{1}{x^4} = +\infty$

2. $\lim_{x \to 0^+} \frac{1}{x^5} = +\infty$

3. $\lim_{x \to 0^+} \frac{1}{x^{21}} = +\infty$

4. $\lim_{x \to 0^-} \frac{1}{x^8} = +\infty$

5. $\lim_{x \to 0^-} \frac{1}{x^{15}} = -\infty$

6. $\lim_{x \to 0^-} \frac{1}{x^{2019}} = -\infty$

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INFINITE LIMITS

Theorem

Suppose $\lim_{x \to c} f(x) = L \neq 0$ and $\lim_{x \to c} g(x) = 0$. If

i. $L > 0$ and $g(x)$ approaches 0 through positive values as x approaches c, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = +\infty$$

ii. $L > 0$ and $g(x)$ approaches 0 through negative values as x approaches c, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = -\infty$$

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INFINITE LIMITS

Theorem

Suppose $\lim_{x \to c} f(x) = L \neq 0$ and $\lim_{x \to c} g(x) = 0$. If

i. $L < 0$ and $g(x)$ approaches 0 through positive values as x approaches c, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = -\infty$$

ii. $L < 0$ and $g(x)$ approaches 0 through negative values as x approaches c, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = +\infty$$

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EXAMPLES

Evaluate the following infinite limits

1. $\lim_{x \to 1^+} \frac{x+1}{x-1} = \frac{2}{0^+} = +\infty$

2. $\lim_{x \to 2^-} \frac{x^2-1}{x-2} = \frac{3}{0^-} = -\infty$

3. $\lim_{x \to 0^-} \frac{2x+7x-5}{x} = \frac{-5}{0^-} = +\infty$

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OTHER EXAMPLES

Evaluate the following infinite limits.

1. $\lim_{x \to 1^+} \frac{2x+3}{x^2-1}$

2. $\lim_{x \to 2^-} \frac{3x^2+x-7}{x^2+3x-10}$

3. $\lim_{x \to -3^-} \frac{x^2+4x-1}{x^2-2x-15}$ =

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TRY THIS!

Evaluate the following infinite limits

1. $\lim_{x \to 0^-} \frac{1}{x^{12}}$

2. $\lim_{x \to 0^-} \frac{1}{x^{23}}$

3. $\lim_{x \to 0^-} \frac{1}{x^{123456}}$

4. $\lim_{x \to 0^-} \frac{x - 100}{x^2 + x}$

5. $\lim_{x \to -1^-} \frac{3x^2 - 27}{x^2 - 1}$

6. $\lim_{x \to -5^+} \frac{x^2 - 125}{x^2 + 4x - 5}$

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LIMITS AT INFINITY

Let $f(x) = \frac{1}{x}$. Observe what happens to $f(x)$-values as x increases without bound.

Since, the $f(x)$-values approach 0 as x goes to infinity, then

$$\lim_{x \to +\infty} f(x) = 0$$

x	$f(x)$
1	**1**
10	**0.1**
100	**0.01**
1,000	**0.001**
10,000	**0.0001**
100,000	**0.00001**
1,000,000	**0.000001**

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LIMITS AT INFINITY

Theorem

If k is any real number and n is positive integer, then

$$\lim_{x \to \infty} k = k$$

$$\lim_{x \to \infty} \frac{1}{x^n} = 0$$

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SAMPLE PROBLEMS

Evaluate each limit:

1. $\lim_{x \to +\infty} (x - 1)$

2. $\lim_{x \to -\infty} (x^2 - 2x + 3)$

3. $\lim_{x \to \infty} \left(\frac{2x^2 - 3x + 1}{x^2 + 4x + 1} \right)$

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SAMPLE PROBLEMS

SOLUTION:

1. $\lim_{x \to +\infty} (x - 1) = +\infty + 1 = +\infty$

2. $\lim_{x \to -\infty} (x^2 - 2x + 3) = (-\infty)^2 - 2(-\infty) + 3$

$= \infty + \infty + 3 = +\infty$

3. $\lim_{x \to \infty} \left(\frac{2x^2 - 3x + 1}{x^2 + 4x + 1} \right) = 2$

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PRACTICE PROBLEMS

Evaluate each limit:

1. $\lim_{x \to -\infty} (x + 10)$

2. $\lim_{x \to -\infty} (x^2 - 4x + 1)$

3. $\lim_{x \to \infty} \left(\frac{3x + 1}{2x^2 + x + 9} \right)$

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